

TITLE: Standardized Anticoagulation Infusion Dosing for VV, VP, and VA ECMO (Adult ECMO Anticoagulation Order Set Updates)

PROTOCOL: ECMO Anticoagulation

PURPOSE: To update and standardize current Adult ECMO anticoagulation protocols and order sets to enhance ease of medication ordering and minimize risk for patient harm. This protocol/order set will standardize anticoagulant goal ranges, minimize manual manipulation to anticoagulation administration instructions within Epic, and minimize the amount of supplemental laboratory orders that must be placed by the ordering provider. Additionally, this protocol aims to incorporate safer monitoring strategies during ECMO support by standardizing laboratory monitoring techniques during ECMO support.

APPROVED:



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DEFINITIONS:

- A. ACT: activated clotting time
- B. aPTT: activated partial thromboplastin time
- C. CBC: complete blood count
- D. ECMO: extracorporeal membrane oxygenation
- E. INR: international normalized ratio
- F. IV: intravenous
- G. POC: point of care
- H. PT: prothrombin time
- I. TEG: thromboelastography
- J. UFH: unfractionated heparin
- K. VA: Veno-arterial
- L. VV: Veno-venous
- M. VP: Veno-pulmonary

CORE PROCEDURE:

Adult ECMO Anticoagulation Orders – GNV “Supplemental”

DO NOT ORDER HEPARIN OR BIVALRUDIN FOR ADULT ECMO OUTSIDE OF THIS ORDER SET

Medications

1. VA ECMO Anticoagulation Options

- a. Heparin Anti-Xa Monitoring Protocols
 - i. High Bleeding Risk (Goal Anti-Xa (unfractionated heparin level) 0.1-0.2)
 - 1. Heparin 25000 units/250 mL
 - 2. Rate: 0-30 units/kg/hr
 - 3. Route: IV
 - 4. Admin instructions: ATTENTION NURSES: Use heparin dosing weight specified in order to program Alaris pump. Once infusion initiated, do not change weight used to program pump (even if order modified). Please contact pharmacy with questions.
 - a. Check unfractionated heparin level every 4 hours after starting infusion for 24 hours, then every 6 hours until ECMO decannulation, unless otherwise specified by ECMO provider. Ensure any recent dose change has been in place for at least 4 hours prior to draw. DO NOT RETIME HEPARIN LEVEL unless timing is greater than one hour off schedule.
 - b. HIGH BLEED RISK PROTOCOL

- c. Initiate infusion at 5 units/kg/hr (use heparin dosing weight).
Goal UFH level is 0.1 to 0.2. Adjust dose based on unfractionated heparin value.
- d. For unfractionated heparin level (units/mL) less than 0.1:
increase by 1 unit/kg/hr (use heparin dosing weight)
- e. For unfractionated heparin level (units/mL) between 0.1 to 0.2: GOAL RANGE, no change
- f. For unfractionated heparin level (units/mL) 0.21 to 0.3:
decrease by 1 unit/kg/hr (use heparin dosing weight)
- g. For unfractionated heparin level (units/mL) 0.31 to 0.4:
decrease by 2 units/kg/hr (use heparin dosing weight)
- h. For unfractionated heparin level (units/mL) 0.41 to 0.5: hold
heparin infusion for 30 minutes and then decrease by 2
units/kg/hr (use heparin dosing weight)
- i. For unfractionated heparin level (units/mL) 0.51 and greater:
hold heparin infusion for 60 minutes and then decrease by 3
units/kg/hr (use heparin dosing weight)
- ii. Low Bleeding Risk (Goal Anti-Xa (unfractionated heparin) 0.2-0.3)
 - 1. Heparin 25000 units/250 mL
 - 2. Rate: 0-30 units/kg/hr
 - 3. Route: IV
 - 4. Admin instructions: ATTENTION NURSES: Use heparin dosing weight
specified in order to program Alaris pump. Once infusion initiated,
do not change weight used to program pump (even if order
modified). Please contact pharmacy with questions.
 - a. Check unfractionated heparin level every 4 hours after
starting infusion for 24 hours, then every 6 hours until ECMO
decannulation, unless otherwise specified by ECMO
provider. Ensure any recent dose change has been in place
for at least 4 hours prior to draw. DO NOT RETIME HEPARIN
LEVEL unless timing is greater than one hour off schedule.
 - b. LOW BLEEDING RISK PROTOCOL
 - c. Initiate infusion at 10 units/kg/hr. Goal UFH level is 0.2 to 0.3.
Adjust dose based on unfractionated heparin value.
 - d. For unfractionated heparin level (units/mL) less than 0.1:
increase by 2 unit/kg/hr (use heparin dosing weight)
 - e. For unfractionated heparin level (units/mL) 0.1 to 0.19:
increase by 1 unit/kg/hr (use heparin dosing weight)
 - f. For unfractionated heparin level (units/mL) between 0.2 to
0.3: GOAL RANGE, no change

- g. For unfractionated heparin level (units/mL) 0.31 to 0.4: decrease by 1 units/kg/hr (use heparin dosing weight)
 - h. For unfractionated heparin level (units/mL) 0.41 to 0.5: hold heparin infusion for 30 minutes and then decrease by 2 units/kg/hr (use heparin dosing weight)
 - i. For unfractionated heparin level (units/mL) 0.51 and greater: hold heparin infusion for 60 minutes and then decrease by 3 units/kg/hr (use heparin dosing weight)
 - iii. PE/DVT/Clot/thrombosis (Goal Anti-Xa (unfractionated heparin) 0.3-0.4)
 - 1. Heparin 25000 units/250 mL
 - 2. Rate: 0-30 units/kg/hr
 - 3. Route: IV
 - 4. Admin instructions: ATTENTION NURSES: Use heparin dosing weight specified in order to program Alaris pump. Once infusion initiated, do not change weight used to program pump (even if order modified). Please contact pharmacy with questions.
 - a. Check unfractionated heparin level every 4 hours after starting infusion for 24 hours, then every 6 hours until ECMO decannulation, unless otherwise specified by ECMO provider. Ensure any recent dose change has been in place for at least 4 hours prior to draw. DO NOT RETIME HEPARIN LEVEL unless timing is greater than one hour off schedule.
 - b. LOW BLEEDING RISK & CLOTS PRESENT PROTOCOL
 - c. Initiate infusion at 12 units/kg/hr. Goal UFH level is 0.3 to 0.4. Adjust dose based on unfractionated heparin value.
 - d. For unfractionated heparin level (units/mL) less than 0.2: increase by 2 unit/kg/hr (use heparin dosing weight)
 - e. For unfractionated heparin level (units/mL) 0.2 to 0.29: increase by 1 unit/kg/hr (use heparin dosing weight)
 - f. For unfractionated heparin level (units/mL) between 0.3 to 0.4: GOAL RANGE, no change
 - g. For unfractionated heparin level (units/mL) 0.41 to 0.5: decrease by 1 unit/kg/hr (use heparin dosing weight)
 - h. For unfractionated heparin level (units/mL) 0.51 to 0.6: hold heparin infusion for 30 minutes and then decrease by 2 units/kg/hr (use heparin dosing weight)
 - i. For unfractionated heparin level (units/mL) 0.61 and greater: hold heparin infusion for 60 minutes and then decrease by 3 units/kg/hr (use heparin dosing weight)
 - iv. Labs for UFH Anti-Xa Protocols:

1. ****FOR HEPARIN Anti-Xa MONITORING ONLY**** UFH level at baseline and then every 4 hours for 24 hours; then every 6 hours until ECMO decannulation
 2. Daily CBC
 3. Daily TEG
 4. Daily Fibrinogen
 5. Daily Anti-Thrombin III level
 6. Daily Plasma-free hemoglobin
 7. PT/INR at baseline, then weekly
- b. Heparin Induced Thrombocytopenia (Argatroban, goal aPTT 50-60)
- i. Argatroban 50 mg in 50 mL
 - ii. Rate: 0-10 mcg/kg/min
 - iii. Route: IV
 - iv. Admin instructions: ATTENTION NURSES: Use argatroban dosing weight specified in order to program Alaris pump. Once infusion initiated, do not change weight used to program pump (even if order modified). Please contact pharmacy with questions.
 1. Check aPTT level every 4 hours after starting argatroban infusion for 24 hours, then every 6 hours until ECMO decannulation. Ensure any recent dose change has been in place for at least 4 hours prior to draw. DO NOT RETIME aPTT LEVEL unless timing is greater than one hour off schedule.
 2. VA ECMO & HIT/HITT DOSING PROTOCOL
 3. Start infusion at 0.2 mcg/kg/min. Goal aPTT is 50-60 sec. Adjust dose based on aPTT value.
 4. For aPTT (sec) below 30, increase dose by 0.2 mcg/kg/min
 5. For aPTT (sec) from 30 to 39, increase dose by 0.15 mcg/kg/min
 6. For aPTT (sec) from 40 to 49, increase dose by 0.1 mcg/kg/min
 7. For aPTT (sec) from 50 to 60, GOAL RANGE do not change dose.
 8. For aPTT (sec) from 61 to 70, decrease dose by 0.1 mcg/kg/min
 9. For aPTT (sec) from 71 to 80, stop infusion for 60 minutes, then restart, decrease dose by 0.15 mcg/kg/min
 10. For aPTT (sec) above 80, stop infusion for 90 minutes, then restart, decrease dose by 0.2 mcg/kg/min
 11. Notify Physician if patient requires greater than argatroban 10 mcg/kg/min without achieving a therapeutic aPTT.
 12. Obtain PT/INR weekly during argatroban infusion therapy. INR will increase with argatroban monotherapy; however, the INR elevation is not a reflection of steady state anticoagulation.

13. Notify Physician if there are active orders for unfractionated heparin (including flushes), dalteparin, enoxaparin, bivalirudin, lepirudin, and/or fondaparinux.
- v. Labs for Argatroban for HIT
 1. ****FOR ARGATROBAN ONLY**** aPTT at baseline and then every 4 hours for 24 hours; then every 6 hours until ECMO decannulation
 2. Daily CBC
 3. Daily TEG
 4. Daily Fibrinogen
 5. Daily Plasma-free hemoglobin
 6. PT/INR at baseline, then weekly
 7. Baseline hepatic function panel
- c. Heparin ACT-based Protocols For VA-ECMO (RESERVED FOR PATIENTS WHERE Anti-Xa MONITORING IS UNRELIABLE) ****ACT-based protocols are reserved for patients with hyperbilirubinemia (Tbili greater than 10 mg/dL), hemolysis (plasma free hemoglobin greater than 200 mg/dL), icterus, and lipemia (triglycerides greater than 600 mg/dL)****
 - i. Clots present in circuit (ACT goal 220-250)
 1. Rate: 0-2000 units/hr
 2. Route: IV
 3. Frequency: Continuous
 4. Administration Instructions:
 - a. ECMO Heparin Infusion: Clots present in the circuit
 - b. Therapeutic ACT goal = 220 - 250 sec
 - c. Initiate heparin infusion at 500 units per hour. Check Activated Clotting Time (ACT) every hour and adjust per instructions below.
 - d. Activated Clotting Time (seconds): Infusion Dose Change
 - i. Less than 180: Increase dose by 300 units/hour
 - ii. 180 - 199: Increase dose by 200 units/hour
 - iii. 200 - 219: Increase dose by 100 units/hour
 - iv. 220 - 250: No dose change (GOAL)
 - v. 251 - 270: Decrease dose by 100 units/hour
 - vi. 271 - 300: Decrease dose by 200 units/hour
 - vii. Greater than 300: Stop infusion for 1 hour, then decrease dose by 200 units/hour
 - e. Notify MD if patient requires greater than 48,000 units / day of IV heparin without reaching therapeutic ACT level
 - f. Therapeutic ACT goal can only be modified by MD or bedside Perfusionist to account for variations in pump flow.

- g. Notify physician if any bleeding occurs, (including increased chest tube output), if platelet count decreases to less than 100,000 cells/mm³ or by 50% from baseline, or if fibrin formation/clots are seen in ECMO circuit.
 - ii. No Clots/Low Bleeding Risk (ACT goal 200-220)
 - 1. Rate: 0-2000 units/hr
 - 2. Route: IV
 - 3. Frequency: Continuous
 - 4. Administration Instructions:
 - a. ECMO Heparin Infusion: No Clots/Low Bleeding Risk
 - b. Therapeutic ACT goal = 200 - 220 sec
 - c. Initiate heparin infusion at 500 units per hour. Check Activated Clotting Time (ACT) every hour and adjust per instructions below.
 - d. Activated Clotting Time (seconds): Infusion Dose Change
 - i. Less than 180: Increase dose by 200 units/hour
 - ii. 180 – 199: Increase dose by 100 units/hour
 - iii. 200 - 220: No dose change (GOAL)
 - iv. 221 - 250: Decrease dose by 100 units/hour
 - v. 251 - 270: Decrease dose by 200 units/hour
 - vi. Greater than 270: Stop infusion for 1 hour, then decrease dose by 200 units/hour
 - e. Notify MD if patient requires greater than 48,000 units / day of IV heparin without reaching therapeutic ACT level
 - f. Therapeutic ACT goal can only be modified by MD or bedside Perfusionist to account for variations in pump flow.
 - g. Notify physician if any bleeding occurs, (including increased chest tube output), if platelet count decreases to less than 100,000 cells/mm³ or by 50% from baseline, or if fibrin formation/clots are seen in ECMO circuit.
- iii. High Bleeding Risk (ACT goal 190-200)
 - 1. Rate: 0-2000 units/hr
 - 2. Route: IV
 - 3. Frequency: Continuous
 - 4. Administration Instructions:
 - a. ECMO Heparin Infusion: High Bleeding Risk
 - b. Therapeutic ACT goal = 190 - 200 sec
 - c. Initiate heparin infusion at 500 units per hour. Check Activated Clotting Time (ACT) every hour and adjust per instructions below.
 - d. Activated Clotting Time (seconds): Infusion Dose Change

- i. Less than 170: Increase dose by 200 units/hour
 - ii. 170 - 189: Increase dose by 100 units/hour
 - iii. 190 - 200: No dose change (GOAL)
 - iv. 201 - 230: Decrease dose by 100 units/hour
 - v. 231 - 260: Decrease dose by 200 units/hour
 - vi. Greater than 260: Stop infusion for 1 hour, then decrease dose by 200 units/hour
 - e. Notify MD if patient requires greater than 48,000 units / day of IV heparin without reaching therapeutic ACT level
 - f. Therapeutic ACT goal can only be modified by MD or bedside Perfusionist to account for variations in pump flow.
 - g. Notify physician if any bleeding occurs, (including increased chest tube output), if platelet count decreases to less than 100,000 cells/mm³ or by 50% from baseline, or if fibrin formation/clots are seen in ECMO circuit.
- iv. Actively Bleeding (ACT goal 180-190)
- 1. Rate: 0-2000 units/hr
 - 2. Route: IV
 - 3. Frequency: Continuous
 - 4. Administration Instructions:
 - a. ECMO Heparin Infusion: Actively Bleeding
 - b. Therapeutic ACT goal = 180 - 190 sec
 - c. Initiate heparin infusion at 500 units per hour. Check Activated Clotting Time (ACT) every hour and adjust per instructions below.
 - d. Activated Clotting Time (seconds): Infusion Dose Change
 - i. Less than 160: Increase dose by 200 units/hour
 - ii. 160 - 179: Increase dose by 100 units/hour
 - iii. 180 - 190: No dose change (GOAL)
 - iv. 191 - 220: Decrease dose by 100 units/hour
 - v. 221 - 250: Decrease dose by 200 units/hour
 - vi. Greater than 250: Stop infusion for 1 hour, then decrease dose by 200 units/hour
 - e. Notify MD if patient requires greater than 48,000 units / day of IV heparin without reaching therapeutic ACT level
 - f. Therapeutic ACT goal can only be modified by MD or bedside Perfusionist to account for variations in pump flow.
 - g. Notify physician if any bleeding occurs, (including increased chest tube output), if platelet count decreases to less than 100,000 cells/mm³ or by 50% from baseline, or if fibrin formation/clots are seen in ECMO circuit.

- v. Heparin BOLUS for platelet administration in ACT protocol
 - 1. Dose 50-1000 units
 - 2. Route: IV
 - 3. Frequency: PRN
 - a. Reason: other, "For Platelet Administration"
 - 4. Administration Instructions: Administer dose equivalent to one-half (50%) of currently running heparin dose per hour prior to platelet administration.
- vi. Labs for ACT-based heparin protocols:
 - 1. ACT battery (POC) at baseline, then Q 1 hr
 - 2. Daily CBC
 - 3. Daily TEG
 - 4. Daily Fibrinogen
 - 5. Daily Anti-Thrombin III level
 - 6. Daily Plasma-free hemoglobin
 - 7. PT/INR at baseline, then weekly

2. VV-/VP-ECMO Anticoagulation Options:

- a. Bivalirudin (preferred anticoagulant for VV-ECMO unless contraindicated)
 - i. Induction/Very High Bleed Risk (aPTT goal 30-40 sec)
 - 1. Bivalirudin 250 mg/ 50 mL
 - 2. Rate: 0-1 mg/kg/hr
 - 3. Route: IV
 - 4. Admin instructions: ATTENTION NURSES: Use bivalirudin dosing weight specified in order to program Alaris pump. Once infusion initiated, do not change weight used to program pump (even if order modified). Please contact pharmacy with questions.
 - a. Check aPTT level every 4 hours after starting infusion for 24 hours, then every 6 hours until ECMO decannulation, unless otherwise directed by ECMO provider. Ensure any recent dose change has been in place for at least 4 hours prior to draw. DO NOT RETIME aPTT LEVEL unless timing is greater than one hour off schedule.
 - b. ANTICOAGULATION INDUCTION/VERY HIGH BLEED RISK DOSING PROTOCOL
 - c. Start infusion at 0.03 mg/kg/hr. Goal aPTT is 30-40 sec. Adjust dose based on aPTT value.
 - d. For aPTT (sec) below 20, increase dose by 10% (multiply current dose by 1.1).
 - e. For aPTT (sec) from 20 to 29, increase dose by 5% (multiply current dose by 1.05).

- f. For aPTT (sec) from 30 to 40, GOAL RANGE, do not change dose.
 - g. For aPTT (sec) from 41 to 50, decrease dose by 10% (multiply current dose by 0.9).
 - h. For aPTT (sec) from 51 to 60, stop infusion for 30 minutes, then restart, decrease dose by 20% (multiply current dose by 0.8).
 - i. For aPTT (sec) above 60, stop infusion for 60 minutes, then restart, decrease dose by 30% (multiply current dose by 0.7).
 - j. Notify Physician if patient requires greater than bivalirudin 1 mg/kg/hr without achieving a therapeutic aPTT.
 - k. Obtain PT/INR weekly during bivalirudin infusion therapy. INR will increase with bivalirudin monotherapy; however, the INR elevation is not a reflection of steady state anticoagulation.
 - l. Notify Physician if there are active orders for unfractionated heparin (including flushes), dalteparin, enoxaparin, argatroban, lepirudin, and/or fondaparinux.
- ii. High Bleeding Risk (aPTT goal 40-50 sec)
- 1. Bivalirudin 250 mg/ 50 mL
 - 2. Rate: 0-1 mg/kg/hr
 - 3. Route: IV
 - 4. Admin instructions: ATTENTION NURSES: Use bivalirudin dosing weight specified in order to program Alaris pump. Once infusion initiated, do not change weight used to program pump (even if order modified). Please contact pharmacy with questions.
 - a. Check aPTT level every 4 hours after starting infusion for 24 hours, then every 6 hours until ECMO decannulation, unless otherwise directed by ECMO provider. Ensure any recent dose change has been in place for at least 4 hours prior to draw. DO NOT RETIME aPTT LEVEL unless timing is greater than one hour off schedule.
 - b. HIGH BLEED RISK DOSING PROTOCOL
 - c. Start infusion at 0.05 mg/kg/hr. Goal aPTT is 40-50 sec. Adjust dose based on aPTT value.
 - d. For aPTT (sec) below 20, increase dose by 30% (multiply current dose by 1.3).
 - e. For aPTT (sec) from 20 to 29, increase dose by 20% (multiply current dose by 1.2).
 - f. For aPTT (sec) from 30 to 39, increase dose by 10% (multiply current dose by 1.1).

- g. For aPTT (sec) from 40 to 50, GOAL RANGE, do not change dose.
- 5. For aPTT (sec) from 51 to 60, decrease dose by 10% (multiply current dose by 0.9).
 - a. For aPTT (sec) from 61 to 70, stop infusion for 30 minutes, then restart, decrease dose by 20% (multiply current dose by 0.8).
 - b. For aPTT (sec) above 70, stop infusion for 60 minutes, then restart, decrease dose by 30% (multiply current dose by 0.7).
 - c. Notify Physician if patient requires greater than bivalirudin 1 mg/kg/hr without achieving a therapeutic aPTT.
 - d. Obtain PT/INR weekly during bivalirudin infusion therapy. INR will increase with bivalirudin monotherapy; however, the INR elevation is not a reflection of steady state anticoagulation.
 - e. Notify Physician if there are active orders for unfractionated heparin (including flushes), dalteparin, enoxaparin, argatroban, lepirudin, and/or fondaparinux.
- iii. Low Bleeding Risk (aPTT goal 50-60 sec)
 - 1. Bivalirudin 250 mg/ 50 mL
 - 2. Rate: 0-1 mg/kg/hr
 - 3. Route: IV
 - 4. Admin instructions: ATTENTION NURSES: Use bivalirudin dosing weight specified in order to program Alaris pump. Once infusion initiated, do not change weight used to program pump (even if order modified). Please contact pharmacy with questions.
 - a. Check aPTT level every 4 hours after starting infusion for 24 hours, then every 6 hours until ECMO decannulation, unless otherwise directed by ECMO provider. Ensure any recent dose change has been in place for at least 4 hours prior to draw. DO NOT RETIME aPTT LEVEL unless timing is greater than one hour off schedule.
 - b. LOW BLEED RISK DOSING PROTOCOL
 - c. Start infusion at 0.05 mg/kg/hr. Goal aPTT is 50-60 sec. Adjust dose based on aPTT value.
 - d. For aPTT (sec) below 30, increase dose by 30% (multiply current dose by 1.3).
 - e. For aPTT (sec) from 30 to 39, increase dose by 20% (multiply current dose by 1.2).
 - f. For aPTT (sec) from 40 to 49, increase dose by 10% (multiply current dose by 1.1).

- g. For aPTT (sec) from 50 to 60, GOAL RANGE, do not change dose.
 - h. For aPTT (sec) from 61 to 70, decrease dose by 10% (multiply current dose by 0.9).
 - i. For aPTT (sec) from 71 to 80, stop infusion for 30 minutes, then restart, decrease dose by 20% (multiply current dose by 0.8).
 - j. For aPTT (sec) above 80, stop infusion for 60 minutes, then restart, decrease dose by 30% (multiply current dose by 0.7).
 - k. Notify Physician if patient requires greater than bivalirudin 1 mg/kg/hr without achieving a therapeutic aPTT.
 - l. Obtain PT/INR weekly during bivalirudin infusion therapy. INR will increase with bivalirudin monotherapy; however, the INR elevation is not a reflection of steady state anticoagulation.
 - m. Notify Physician if there are active orders for unfractionated heparin (including flushes), dalteparin, enoxaparin, argatroban, lepirudin, and/or fondaparinux.
- iv. PE/DVT/Clot/thrombosis (aPTT goal 60-80 sec)
- 1. Bivalirudin 250 mg/ 50 mL
 - 2. Rate: 0-1 mg/kg/hr
 - 3. Route: IV
 - 4. Admin instructions: ATTENTION NURSES: Use bivalirudin dosing weight specified in order to program Alaris pump. Once infusion initiated, do not change weight used to program pump (even if order modified). Please contact pharmacy with questions.
 - a. Check aPTT level every 4 hours after starting infusion for 24 hours, then every 6 hours until ECMO decannulation, unless otherwise directed by ECMO provider. Ensure any recent dose change has been in place for at least 4 hours prior to draw. DO NOT RETIME aPTT LEVEL unless timing is greater than one hour off schedule.
 - b. LOW BLEED RISK DOSING PROTOCOL
 - c. Start infusion at 0.05 mg/kg/hr. Goal aPTT is 60-80 sec. Adjust dose based on aPTT value.
 - d. For aPTT (sec) below 40, increase dose by 30% (multiply current dose by 1.3).
 - e. For aPTT (sec) from 40 to 49, increase dose by 20% (multiply current dose by 1.2).
 - f. For aPTT (sec) from 50 to 59, increase dose by 10% (multiply current dose by 1.1).

- g. For aPTT (sec) from 60 to 80, GOAL RANGE, do not change dose.
- 5. For aPTT (sec) from 81 to 90, decrease dose by 10% (multiply current dose by 0.9).
 - a. For aPTT (sec) from 91 to 100, stop infusion for 30 minutes, then restart, decrease dose by 20% (multiply current dose by 0.8).
 - b. For aPTT (sec) above 100, stop infusion for 60 minutes, then restart, decrease dose by 30% (multiply current dose by 0.7).
 - c. Notify Physician if patient requires greater than bivalirudin 1 mg/kg/hr without achieving a therapeutic aPTT.
 - d. Obtain PT/INR weekly during bivalirudin infusion therapy. INR will increase with bivalirudin monotherapy; however, the INR elevation is not a reflection of steady state anticoagulation.
 - e. Notify Physician if there are active orders for unfractionated heparin (including flushes), dalteparin, enoxaparin, argatroban, lepirudin, and/or fondaparinux.
- v. Labs for bivalirudin protocols:
 - 1. FOR BIVALIRUDIN** aPTT at baseline and then every 4 hours for 24 hours; then every 6 hours until ECMO decannulation
 - 2. Hepatic function panel at baseline
 - 3. Daily CBC
 - 4. Daily TEG
 - 5. Daily Fibrinogen
 - 6. Daily Anti-Thrombin III level
 - 7. Daily Plasma-free hemoglobin
 - 8. PT/INR at baseline, then weekly
- b. Heparin Anticoagulation Choices (reserved for pregnant patients or others unable to use DTI)
 - i. Anti-Xa Protocols (Preferred):
 - 1. High Bleeding Risk (Goal Anti-Xa (unfractionated heparin level) 0.1-0.2)
 - a. Heparin 25000 units/250 mL
 - b. Rate: 0-30 units/kg/hr
 - c. Route: IV
 - d. Admin instructions: ATTENTION NURSES: Use heparin dosing weight specified in order to program Alaris pump. Once infusion initiated, do not change weight used to program pump (even if order modified). Please contact pharmacy with questions.

- i. Check unfractionated heparin level every 4 hours after starting infusion for 24 hours, then every 6 hours until ECMO decannulation. Ensure any recent dose change has been in place for at least 4 hours prior to draw. DO NOT RETIME HEPARIN LEVEL unless timing is greater than one hour off schedule.
 - ii. HIGH BLEED RISK PROTOCOL
 - iii. Initiate infusion at 5 units/kg/hr (use heparin dosing weight). Goal UFH level is 0.1 to 0.2. Adjust dose based on unfractionated heparin value.
 - iv. For unfractionated heparin level (units/mL) less than 0.1: increase by 1 unit/kg/hr (use heparin dosing weight)
 - v. For unfractionated heparin level (units/mL) between 0.1 to 0.2: GOAL RANGE, no change
 - vi. For unfractionated heparin level (units/mL) 0.21 to 0.3: decrease by 1 unit/kg/hr (use heparin dosing weight)
 - vii. For unfractionated heparin level (units/mL) 0.31 to 0.4: decrease by 2 units/kg/hr (use heparin dosing weight)
 - viii. For unfractionated heparin level (units/mL) 0.41 to 0.5: hold heparin infusion for 30 minutes and then decrease by 2 units/kg/hr (use heparin dosing weight)
 - ix. For unfractionated heparin level (units/mL) 0.51 and greater: hold heparin infusion for 60 minutes and then decrease by 3 units/kg/hr (use heparin dosing weight)
2. Low Bleeding Risk (Goal Anti-Xa (unfractionated heparin) 0.2-0.3)
- a. Heparin 25000 units/250 mL
 - b. Rate: 0-30 units/kg/hr
 - c. Route: IV
 - d. Admin instructions: ATTENTION NURSES: Use heparin dosing weight specified in order to program Alaris pump. Once infusion initiated, do not change weight used to program pump (even if order modified). Please contact pharmacy with questions.
 - i. Check unfractionated heparin level every 4 hours after starting infusion for 24 hours, then every 6 hours until ECMO decannulation. Ensure any recent dose change has been in place for at least 4 hours prior to

draw. DO NOT RETIME HEPARIN LEVEL unless timing is greater than one hour off schedule.

ii. LOW BLEEDING RISK PROTOCOL

- iii. Initiate infusion at 10 units/kg/hr. Goal UFH level is 0.2 to 0.3. Adjust dose based on unfractionated heparin value.
- iv. For unfractionated heparin level (units/mL) less than 0.1: increase by 2 unit/kg/hr (use heparin dosing weight)
- v. For unfractionated heparin level (units/mL) 0.1 to 0.19: increase by 1 unit/kg/hr (use heparin dosing weight)
- vi. For unfractionated heparin level (units/mL) between 0.2 to 0.3: GOAL RANGE, no change
- vii. For unfractionated heparin level (units/mL) 0.31 to 0.4: decrease by 1 units/kg/hr (use heparin dosing weight)
- viii. For unfractionated heparin level (units/mL) 0.41 to 0.5: hold heparin infusion for 30 minutes and then decrease by 2 units/kg/hr (use heparin dosing weight)
- ix. For unfractionated heparin level (units/mL) 0.51 and greater: hold heparin infusion for 60 minutes and then decrease by 3 units/kg/hr (use heparin dosing weight)

3. PE/DVT/Clot/thrombosis (Goal Anti-Xa (unfractionated heparin) 0.3-0.4)

- a. Heparin 25000 units/250 mL
- b. Rate: 0-30 units/kg/hr
- c. Route: IV
- d. Admin instructions: ATTENTION NURSES: Use heparin dosing weight specified in order to program Alaris pump. Once infusion initiated, do not change weight used to program pump (even if order modified). Please contact pharmacy with questions.
 - i. Check unfractionated heparin level every 4 hours after starting infusion for 24 hours, then every 6 hours until ECMO decannulation. Ensure any recent dose change has been in place for at least 4 hours prior to draw. DO NOT RETIME HEPARIN LEVEL unless timing is greater than one hour off schedule.
 - ii. LOW BLEEDING RISK & CLOTS PRESENT PROTOCOL

- iii. Initiate infusion at 12 units/kg/hr. Goal UFH level is 0.3 to 0.4. Adjust dose based on unfractionated heparin value.
- iv. For unfractionated heparin level (units/mL) less than 0.2: increase by 2 unit/kg/hr (use heparin dosing weight)
- v. For unfractionated heparin level (units/mL) 0.2 to 0.29: increase by 1 unit/kg/hr (use heparin dosing weight)
- vi. For unfractionated heparin level (units/mL) between 0.3 to 0.4: GOAL RANGE, no change
- vii. For unfractionated heparin level (units/mL) 0.41 to 0.5: decrease by 1 unit/kg/hr (use heparin dosing weight)
- viii. For unfractionated heparin level (units/mL) 0.51 to 0.6: hold heparin infusion for 30 minutes and then decrease by 2 units/kg/hr (use heparin dosing weight)
- ix. For unfractionated heparin level (units/mL) 0.61 and greater: hold heparin infusion for 60 minutes and then decrease by 3 units/kg/hr (use heparin dosing weight)

4. Anti-Xa Monitoring Protocol Labs:

- a. ****FOR HEPARIN Anti-Xa MONITORING ONLY**** UFH level at baseline and then every 4 hours for 24 hours; then every 6 hours until ECMO decannulation
 - b. Daily CBC
 - c. Daily TEG
 - d. Daily Fibrinogen
 - e. Daily Anti-Thrombin III level
 - f. Daily Plasma-free hemoglobin
 - g. PT/INR at baseline, then weekly
- ii. Heparin ACT-based Protocols For VV-ECMO (RESERVED FOR PATIENTS WHERE Anti-Xa MONITORING IS UNRELIABLE) ****ACT-based protocols are reserved for patients with hyperbilirubinemia (Tbili greater than 10 mg/dL), hemolysis (plasma free hemoglobin greater than 200 mg/dL), icterus, and lipemia (triglycerides greater than 600 mg/dL)****
- 1. Clots present in circuit (ACT goal 220-250)
 - a. Rate: 0-2000 units/hr
 - b. Route: IV
 - c. Frequency: Continuous
 - d. Administration Instructions:

- i. ECMO Heparin Infusion: Clots present in the circuit
 - ii. Therapeutic ACT goal = 220 - 250 sec
 - iii. Initiate heparin infusion at 500 units per hour. Check Activated Clotting Time (ACT) every hour and adjust per instructions below.
 - iv. Activated Clotting Time (seconds): Infusion Dose Change
 - II. Less than 180: Increase dose by 300 units/hour
 - III. 180 - 199: Increase dose by 200 units/hour
 - IV. 200 - 219: Increase dose by 100 units/hour
 - V. 220 - 250: No dose change (GOAL)
 - VI. 251 - 270: Decrease dose by 100 units/hour
 - VII. 271 - 300: Decrease dose by 200 units/hour
 - VIII. Greater than 300: Stop infusion for 1 hour, then decrease dose by 200 units/hour
 - v. Notify MD if patient requires greater than 48,000 units / day of IV heparin without reaching therapeutic ACT level
 - vi. Therapeutic ACT goal can only be modified by MD or bedside Perfusionist to account for variations in pump flow.
 - vii. Notify physician if any bleeding occurs, (including increased chest tube output), if platelet count decreases to less than 100,000 cells/mm³ or by 50% from baseline, or if fibrin formation/clots are seen in ECMO circuit.
2. No Clots/Low Bleeding Risk (ACT goal 200-220)
- a. Rate: 0-2000 units/hr
 - b. Route: IV
 - c. Frequency: Continuous
 - d. Administration Instructions:
 - i. ECMO Heparin Infusion: No Clots/Low Bleeding Risk
 - ii. Therapeutic ACT goal = 200 - 220 sec
 - iii. Initiate heparin infusion at 500 units per hour. Check Activated Clotting Time (ACT) every hour and adjust per instructions below.

- iv. Activated Clotting Time (seconds): Infusion Dose Change
 - II. Less than 180: Increase dose by 200 units/hour
 - III. 180 - 199: Increase dose by 100 units/hour
 - IV. 200 - 220: No dose change (GOAL)
 - V. 221 - 250: Decrease dose by 100 units/hour
 - VI. 251 - 270: Decrease dose by 200 units/hour
 - VII. Greater than 270: Stop infusion for 1 hour, then decrease dose by 200 units/hour
 - v. Notify MD if patient requires greater than 48,000 units / day of IV heparin without reaching therapeutic ACT level
 - vi. Therapeutic ACT goal can only be modified by MD or bedside Perfusionist to account for variations in pump flow.
 - vii. Notify physician if any bleeding occurs, (including increased chest tube output), if platelet count decreases to less than 100,000 cells/mm³ or by 50% from baseline, or if fibrin formation/clots are seen in ECMO circuit.
3. High Bleeding Risk (ACT goal 190-200)
- a. Rate: 0-2000 units/hr
 - b. Route: IV
 - c. Frequency: Continuous
 - d. Administration Instructions:
 - i. ECMO Heparin Infusion: High Bleeding Risk
 - ii. Therapeutic ACT goal = 190 - 200 sec
 - iii. Initiate heparin infusion at 500 units per hour. Check Activated Clotting Time (ACT) every hour and adjust per instructions below.
 - iv. Activated Clotting Time (seconds): Infusion Dose Change
 - II. Less than 170: Increase dose by 200 units/hour
 - III. 170 - 189: Increase dose by 100 units/hour
 - IV. 190 - 200: No dose change (GOAL)

- V. 201 - 230: Decrease dose by 100 units/hour
 - VI. 231 - 260: Decrease dose by 200 units/hour
 - VII. Greater than 260: Stop infusion for 1 hour, then decrease dose by 200 units/hour
 - v. Notify MD if patient requires greater than 48,000 units / day of IV heparin without reaching therapeutic ACT level
 - vi. Therapeutic ACT goal can only be modified by MD or bedside Perfusionist to account for variations in pump flow.
 - vii. Notify physician if any bleeding occurs, (including increased chest tube output), if platelet count decreases to less than 100,000 cells/mm³ or by 50% from baseline, or if fibrin formation/clots are seen in ECMO circuit.
4. Actively Bleeding (ACT goal 180-190)
- a. Rate: 0-2000 units/hr
 - b. Route: IV
 - c. Frequency: Continuous
 - d. Administration Instructions:
 - i. ECMO Heparin Infusion: Actively Bleeding
 - ii. Therapeutic ACT goal = 180 - 190 sec
 - iii. Initiate heparin infusion at 500 units per hour. Check Activated Clotting Time (ACT) every hour and adjust per instructions below.
 - iv. Activated Clotting Time (seconds): Infusion Dose Change
 - II. Less than 160: Increase dose by 200 units/hour
 - III. 160 - 179: Increase dose by 100 units/hour
 - IV. 180 - 190: No dose change (GOAL)
 - V. 191 - 220: Decrease dose by 100 units/hour
 - VI. 221 - 250: Decrease dose by 200 units/hour
 - VII. Greater than 250: Stop infusion for 1 hour, then decrease dose by 200 units/hour

- v. Notify MD if patient requires greater than 48,000 units / day of IV heparin without reaching therapeutic ACT level
 - vi. Therapeutic ACT goal can only be modified by MD or bedside Perfusionist to account for variations in pump flow.
 - vii. Notify physician if any bleeding occurs, (including increased chest tube output), if platelet count decreases to less than 100,000 cells/mm³ or by 50% from baseline, or if fibrin formation/clots are seen in ECMO circuit.
5. VV-ECMO (ACT goal 181-200)
- a. Rate: 0-2000 units/hr
 - b. Route: IV
 - c. Frequency: Continuous
 - d. Administration Instructions:
 - i. ECMO Heparin Infusion: Veno-venous (VV) Cannulation
 - ii. Therapeutic ACT goal = 160 - 180 sec
 - iii. Initiate heparin infusion at 500 units per hour. Check Activated Clotting Time (ACT) every hour and adjust per instructions below.
 - iv. Activated Clotting Time (seconds): Infusion Dose Change
 - II. Less than 140: Increase dose by 200 units/hour
 - III. 140 - 159: Increase dose by 100 units/hour
 - IV. 160 - 180: No dose change (GOAL)
 - V. 181 - 200: Decrease dose by 100 units/hour
 - VI. 201 - 230: Decrease dose by 200 units/hour
 - VII. Greater than 230: Stop infusion for 1 hour, then decrease dose by 200 units/hour
 - v. Notify MD if patient requires greater than 48,000 units / day of IV heparin without reaching therapeutic ACT level
 - vi. Therapeutic ACT goal can only be modified by MD to account for variations in pump flow.

- vii. Notify physician if any bleeding occurs, (including increased chest tube output), if platelet count decreases to less than 100,000 cells/mm³ or by 50% from baseline, or if fibrin formation/clots are seen in ECMO circuit.
- 6. Heparin BOLUS for platelet administration in ACT protocol
 - a. Dose 50-1000 units
 - b. Route: IV
 - c. Frequency: PRN
 - i. Reason: other, "For Platelet Administration"
 - d. Administration Instructions: Administer dose equivalent to one-half (50%) of currently running heparin dose per hour prior to platelet administration.
- 7. Labs for ACT-based monitoring protocols
 - a. Daily CBC
 - b. Daily TEG
 - c. Daily Fibrinogen
 - d. Daily Anti-Thrombin III level
 - e. Daily Plasma-free hemoglobin
 - f. PT/INR at baseline, then weekly
 - g. Baseline aPTT
 - h. ACT battery (POC) at baseline, then Q 1 hr

All Anticoagulation Protocols will need: Nursing "Notify" orders - borrowed from "Adult ECMO ACT Based Heparin Infusion Protocol" and other anticoagulation order sets

- Notify physician/provider if bleeding occurs (including increased chest tube output), if platelet count decreases to less than 100,000 cells/mm³ or by 50% from baseline, or if fibrin formation/clots are seen in the ECMO circuit
- Notify physician to d/c other anticoagulants if there are active orders for alternative anticoagulants including intravenous or subcutaneous heparin, heparin flushes, enoxaparin, dalteparin, argatroban, bivalirudin, lepirudin, apixaban, rivaroxaban, dabigatran, edoxaban, and/or fondaparinux.
- Measure weight daily

All Anticoagulation Protocols will need: Precautions - borrowed from "Adult ECMO ACT Based Heparin Infusion Protocol" and other anticoagulation order sets

- Bleeding/thrombocytopenia precautions

ASSOCIATED POLICIES:

- 1) ECMO-PROC-35
- 2) Pharmacy Policy 09-05. Anticoagulation Therapy Monitoring

KEYWORDS:

ECMO, Anticoagulation, Heparin, Bivalirudin, Argatroban

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